Lab. Assignment-11 (b)

Computer Architecture (CS F342)
Semester-I, 2026-27
Department of Computer Science and Information Systems (CSIS)
BITS-Pilani, K K Birla Goa Campus, Goa, India.

Due date: Nov 12, 2026: 1:59 AM Marks: 5

1. Download the *CA_Lab11b_2LevelBP_Template.zip*. Three branch trace files are: *btrace1.txt, btrace2.txt, btrace3.txt.* 2-level branch predictor is available in *2LevelBP.cpp.*

The trace files contain the program's branches' trace. The format is like this: <branch's position in the program/PC values> <T/N>. *T* indicates branch has taken and *N* indicates branch has not taken. 32-bits hexadecimal format is used to represent the PC value. The branch's target address is not present in the trace file. The entire PC's value is stored in the BTB. The BTB consists of eight entries. PC's last 3-bits are used as the index. For the two-level branch predictor, the LHT used PC's last 3-bits as the index. All the modules available in the template file have to be used. No need to define other variables and to modify the other module except the below in the program.

Module	Marks
updateLHT2Level()	1
twoLevelBranchPredictor()	4

The expected outputs are:
For btrace1.txt,

```
PS D:\CSF342-CA\Sem-I-AY-2022-23\Labs\Lab-7-on-29-Nov\2-level Branch Predictor\Sol> .\a.exe .\btrace1.txt
################################################################################
2-level Branch Predictor's Statistics
================================================================================
1 predictor's (68) total prediction: 10001
1 predictor's (68) correct prediction: 9997
1 predictor's (68) accuracy: 0.9996
================================================================================
2 predictor's (108) total prediction: 260000
2 predictor's (108) correct prediction: 249993
2 predictor's (108) accuracy: 0.961512
################################################################################
2-level BP's Average accuracy= 0.980556
```

For btrace2.txt,

```
PS D:\CSF342-CA\Sem-I-AY-2022-23\Labs\Lab-7-on-29-Nov\2-level Branch Predictor\Sol> .\a.exe .\btrace2.txt
#########################################################################
2-level Branch Predictor's Statistics
=========================================================================
1 predictor's (68) total prediction: 10001
1 predictor's (68) correct prediction: 9995
1 predictor's (68) accuracy: 0.9994
=========================================================================
2 predictor's (108) total prediction: 10000
2 predictor's (108) correct prediction: 9998
2 predictor's (108) accuracy: 0.9998
=========================================================================
3 predictor's (124) total prediction: 10000
3 predictor's (124) correct prediction: 10000
3 predictor's (124) accuracy: 1
#########################################################################
2-level BP's Average accuracy= 0.999733
```

For btrace3.txt,

```
PS D:\CSF342-CA\Sem-I-AY-2022-23\Labs\Lab-7-on-29-Nov\2-level Branch Predictor\Sol> .\a.exe .\btrace3.txt
#########################################################################
2-level Branch Predictor's Statistics
=========================================================================
1 predictor's (68) total prediction: 3336
1 predictor's (68) correct prediction: 3331
1 predictor's (68) accuracy: 0.998501
=========================================================================
2 predictor's (108) total prediction: 3335
2 predictor's (108) correct prediction: 3335
2 predictor's (108) accuracy: 1
=========================================================================
3 predictor's (124) total prediction: 3335
3 predictor's (124) correct prediction: 3334
3 predictor's (124) accuracy: 0.9997
=========================================================================
4 predictor's (136) total prediction: 3335
4 predictor's (136) correct prediction: 3329
4 predictor's (136) accuracy: 0.998201
#########################################################################
2-level BP's Average accuracy= 0.999101
```

Download the *CA_Lab-11.zip* and rename the folder as *<CampusID>_ CA_Lab-11*. Output must be stored as *<CampusID>_ CA_Lab-11b_opbt1.png*, *<CampusID>_ CA_Lab11-b_opbt2.png* and *<CampusID>_CA_Lab-11b_opbt3.png* in the folder, 2LevelBP. Create a zip file *<CampusID>_ CA_Lab-11.zip* which contains all the files mention above and submit it in Quanta. Do not create archives in other formats (rar, tar.gz etc). Once uploaded on Quanta, remember to submit for grading. Do not leave it as a draft.

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Note: Please don't upload the assignments, template file/solution and lab. manual on GitHub or others public repository. Kindly remove them, if you have uploaded the previous assignments.

It violates the BITS's Intellectual Property Rights (IPR).

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